

Nicola Muca Cirone

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EDUCATION	PhD Mathematics - CDT Mathematics of Random Systems, 2022-2025 Imperial College London - University of Oxford • Deep Learning, Mathematical Finance, Rough Paths Analysis. • Supervised by <i>Dr. Cristopher Salvi</i>. • Researching scaling limits of Neural Networks. • Developing new Kernel Methods for Time Series analysis. MSc Mathematics, University of Pisa - ETH Zürich, 2021-2022 • Honour's degree (110/110 cum Laude) GPA 30 e Lode • Stochastic Calculus, Stochastic PDEs, Mathematical Finance, ML in Finance. • Thesis: <i>Kernel Methods on Path Spaces</i> supervised by <i>Prof. Josef Teichmann</i>. BSc Mathematics, University of Pisa, 2018-2021 • Honour's degree (110/110 cum Laude) GPA 29.88/30 • Thesis on advanced topics in Geometrical Model Theory.	
PUBLICATIONS	[3] Nicola Muca Cirone, Jad Hamdan and Cristopher Salvi. Graph Expansions of Deep Neural Networks and their Universal Scaling Limits. <i>Submitted to Journal</i> url: https://arxiv.org/abs/2407.08459 [2] Nicola Muca Cirone, Antonio Orvieto, Benjamin Walker, Cristopher Salvi and Terry Lyons. Theoretical Foundations of Deep Selective State-Space Models. <i>Accepted to NeurIPS 2024</i> url: https://arxiv.org/abs/2402.19047 [1] Nicola Muca Cirone, Maud Lemerrier and Cristopher Salvi. Neural signature kernels as infinite-width-depth-limits of controlled ResNets. <i>Proceedings of the 40th International Conference on Machine Learning</i>, 25358–25425, 2023. url: https://proceedings.mlr.press/v202/muca-cirone23a.html	
TALKS	Stochastic Analysis Seminar, ShanghaiTech University, Nov 2024 Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i> DataSig - Rough Paths Interest Group, Alan Turing Institute, May 2024 Title: <i>Signature Reconstruction from Randomized Signatures.</i> Computer Vision Group Seminar, University of Michigan, April 2024 Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i> Talks in Financial and Insurance Mathematics, ETH Zürich, Dec 2023 Title: <i>Neural Signature Kernels (and Trees).</i> 7th London-Paris Bachelier Workshop, Imperial College London, Sept 2023 Title: <i>Rough Kernel Hedging.</i> CDT Mathematics of Random Systems Workshop, University of Oxford, June 2023 Title: <i>Neural Signature Kernels.</i> 16th Oxford-Berlin Young Researchers Meeting, University of Oxford, Dec 2022 Title: <i>Signatures and the infinite-width-depth limit of Data Driven ResNets.</i>	

EXPERIENCE	Industry Project - Neural Signature Kernels , Imperial College London	June 2024
	Co-organized 2024 CDT Industry Project and supervised PhD students in applying Neural Signature Kernels to time series analysis.	
	Teaching Assistant - Stochastic Calculus for Finance , ICL Business School	Oct - Dec 2023
	<i>MSc Risk Management & Financial Engineering</i> . Taught exercise classes.	
	Industry Project - Brain signal analysis with Signatures , Imperial College London	June 2023
	Challenge: analyze MEG data from several subjects, detect the one who had a seizure. Our team used Signature techniques, the success and clarity of our results won the competition.	
	Teaching Assistant - Numerical Methods for Finance , Imperial College London	Feb 2023
	<i>MSc Mathematics and Finance</i> . Taught exercise classes.	
	Teaching Assistant - Data Analysis , Università di Pisa	2021-2022
	<i>MSc Mathematics</i> . Provided tutoring support.	
	NLP Project - Classification of mathematical papers , Università di Pisa	2021-2022
	Task: classify which mathematical subjects an academic paper belongs to, given its title and abstract. The implementation was done in Python.	
AWARDS	NeurIPS 2024 Scholar Award , Neural Information Processing Systems Foundation	2024
	INDAM Scholarship , Istituto Nazionale di Alta Matematica	2022
	Two year scholarship from the Italian Institute of Higher Mathematics. Placed 2nd at National level.	
COMPETENCES	Languages Italian (<i>native</i>), English (C1), French (<i>basic</i>)	
	Technical Skills Python, Jax, Pytorch, TensorFlow, R, Matlab, Java	